

Lab #1 - Introduction, Safety and Measurements

ECE 482/582, Fall 2003

Goals:

1. To understand the safety issues and hazards of working in the optical electronics laboratory.
2. To develop an appreciation for the art and science of taking scientific measurements.
3. Develop an understanding of the theory and application of “Lock-in Amplifiers”.

Reading:

Laser Safety

Kuhn, *Laser Engineering*, pp.441-449

<http://www.dermatology.org/laser/eyesafety.html>

Rule #1 - DO NOT LOOK INTO A LASER BEAM!

Lock-In Amplifiers

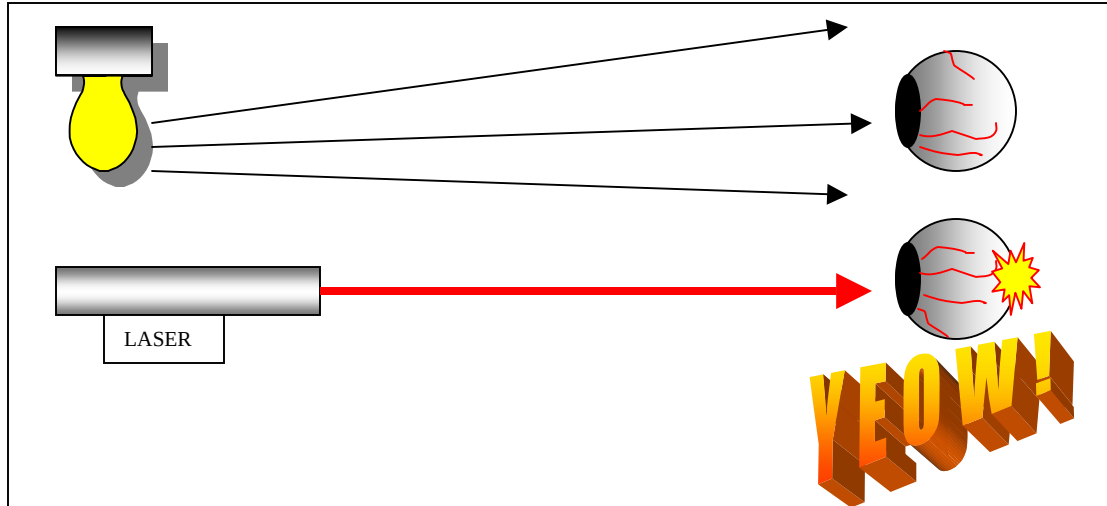
<http://www.cpm.uncc.edu/programs/lia.pdf>

<http://www.cpm.uncc.edu/programs/tn1000.pdf>

Preparatory Questions:

1. What are the three main safety hazards in the Optical Electronics Laboratory?
 - a.
 - b.
 - c.
2. What can you do to prevent accidents from these 3 hazards?
 - a.
 - b.
 - c.
3. You should be familiar with Rule #1 above. Lets compare a 60Watt light bulb to a 5mW laser pointer. It might seem that the laser pointer is 12,000 times less damaging than the light bulb. However, you can stare at a 60Watt light bulb from 1 meter away with little more discomfort than an after-image. If you stare at a 5mW laser pointer too long, your vision will be permanently damaged. The laser

is more damaging because it is a narrow confined beam and your eye can gather most of the emitted energy. However, the light from the light bulb spreads out in all directions and your eye intercepts only a small fraction of the emitted energy. The eye also focuses the laser light to a smaller spot size on the retina than it does with the light from the light bulb (see Kuhn Eq. 4.65).



Example 1: Staring at the 60Watt light bulb at a distance of 1 meter.
The light is spread equally over a sphere of radius 1 meter.

$$SurfaceArea = 4\pi(1m)^2 = 12.6m^2$$

$$CorneaIntensity = \frac{LightBulbPower}{SurfaceArea} = \frac{60Watts}{12.6m^2} = 4.8W / m^2$$

The light will enter through the pupil, which is about 2mm in diameter.

$$PupilArea = \pi(1mm)^2 = 3.1 \times 10^{-6} m^2$$

The total power entering the eye is,

$$EyePower = CorneaIntensity \times PupilArea = 1.5 \times 10^{-5} W$$

The eye focuses the image of the light bulb onto the retina to a spot about 300um in diameter. Therefore, the intensity at the retina is,

$$RetinaIntensity = \frac{EyePower}{SpotSize} = \frac{1.5 \times 10^{-5} W}{\pi(150\mu m)^2} = 212W / m^2$$

Question: Calculate the retinal intensity for a 5mW laser beam entering the eye with a collimated beam diameter of 2mm. The eye will focus this laser beam to a spot diameter of 10µm.

NOTE You will be using lasers that can cause irreparable damage to the human eye. You must be alert at all times and use proper safety techniques while working in the lab. Anyone intentionally misusing a laser will be barred from this laboratory and receive a grade of F.**

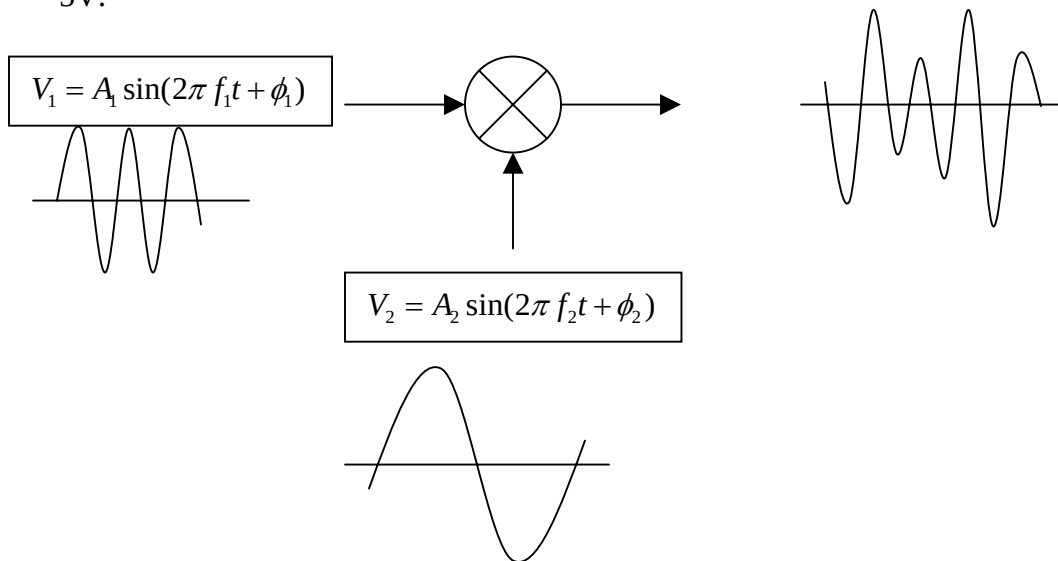
- Describe in basic terms how a “lock in amplifier” works and what kind of signal is needed for its use?

Experiment:

Using the PIN photodiode detector demonstrated in the lab and an oscilloscope, make a measurement of the detected fluorescent lighting in the lab room. Sketch the signal or signals you measure. What is the frequency of the observed signal or signals?

Questions to Answer:

- All lock-in amplifiers use a PSD (phase sensitive detector) that consists of nothing more than a mixer followed by a low-pass filter. Consider the mixer represented in the diagram below. Calculate an expression for the output. Use a math software package of your choice to plot the output of this mixer circuit for $f_1 = f_2 = 30\text{Hz}$ and $\phi_1 = \phi_2 = 0$ radians. Now calculate the output if $\phi_1 = 90$ radians and $\phi_2 = 0$ radians. Plot this result also for $f_1 = f_2 = 30\text{Hz}$ and $A_1 = A_2 = 5\text{V}$.



2. What was the frequency observed with the photodetector when looking at the room lights? Give an explanation for this. Explain how a fluorescent light works (look it up).
3. (Extra Credit) Design a simple low pass filter circuit that has a bandwidth of 30 Hz and drops 6dB/octave (amplitude decreases by $\frac{1}{2}$ for every doubling of frequency). How can I change this circuit to get a drop of 12dB/octave?

LASER SAFETY RULES ECE 482/582 Fall 2003

The following rules will help prevent any accidents in the laboratory when working with lasers. If the safety of a device or object is in question, ask the instructor or lab TA for assistance.

1. Be aware of all operating lasers and electronic equipment in the laboratory.
2. Notify others in the laboratory before approaching or operating laboratory equipment.
3. Do not look into a laser beam no matter what the circumstances.
4. When working with laser beams control and confine them.
5. Be aware of any stray reflections or reflective surfaces that the laser beam may encounter.
6. Do not wear a watch or jewelry with reflective surfaces.
7. Keep laser beams in a horizontal plane below eye level.
8. Keep your eyes above the level of the laser beams.
9. Leave the room light on when possible. This helps you to see what you are working on. Also, the eye pupils open wider when it is dark, presenting a larger target for the laser beam.
10. Wear the appropriate laser safety goggles.
11. Turn off power supplies before working with laser electrical contacts.
12. Use common sense and be alert.
13. Respect both lasers and electricity at all times.